PAPER ID-410959

Subject Code: MNIO401

Roll No: | | | | | | | | | | | | | |

BTECH-MIN

(SEM IV) THEORY EXAMINATION 2023-24

INTRODUCTION TO INTERNET OF THINGS

TIME: 3 HRS **M.MARKS: 100**

Note: 1. Attempt all Sections. If require any missing data; then choose suitably.

SECTION A

1. Attempt *all* questions in brief. **2 x 10 = 20**

Q no.	Question	Marks	CO
a.	How is the convergence of IoT and Web Technology influencing technological advancements and societal impacts?	02	
b.	Explain the vision of the Internet of Things.	02	
c.	How does the international driven global value chain influence the development and deployment of IoT solutions?	02	
d.	Describe the architectural overview from M2M to IoT.	02	
e.	What are the key considerations for deploying IoT systems?	02	
f.	Discuss the significance of a reference model in IoT.	02	
g.	How does IoT enable the transformation of traditional factories into smart factories?	02	
h.	Discuss the role of IoT in the oil and gas industry.	02	
i.	Discuss the security challenges in IoT.	02	
j.	How does trust influence the adoption and success of IoT solutions?	02	

SECTION B

2. Attempt any *three* of the following: **3 x 10 = 30**

a.	Discuss the role of networks and communication processes in the IoT ecosystem. How do these elements contribute to the overall functionality and efficiency of IoT systems?	10	
b.	Discuss the significance of standards considerations in the IoT architecture. How do these standards facilitate interoperability and scalability in IoT systems?	10	
c.	Discuss the information view of the IoT Reference Architecture. How does it facilitate data management and communication within IoT systems?	10	
d.	How do IoT solutions enhance the efficiency and customer experience in retail settings?	10	
e.	Evaluate the contributions from FP7 projects towards addressing IoT privacy, security, and governance challenges. What are the key findings and recommendations from these projects?	10	

SECTION C

3. Attempt any *one* part of the following: **1 x 10 = 10**

a.	Critically assess the current state of IoT-related standardization. What are the existing standards, and what further standardization is required to support the growth of IoT?	10	
b.	Discuss the recommendations on research topics in the field of IoT. How do these recommendations shape the future directions of IoT research and development?	10	

QP24EP2_311 | 24-Jul-2024 1:37:11 PM | 115.245.67.130

QP24EP2_311 | 24-Jul-2024 1:37:11 PM | 115.245.67.130

PAPER ID-410959

Roll No: | | | | | | | | | | | | | |

BTECH-MIN
(SEM IV) THEORY EXAMINATION 2023-24
INTRODUCTION TO INTERNET OF THINGS

TIME: 3 HRS **M.MARKS: 100**

4. Attempt any *one* part of the following: 1 x 10 = 10

a.	Explain the concept of an IoT architecture outline. How does it address the key design principles and needed capabilities for IoT systems?	10	
b.	Discuss the evolution from M2M to IoT in terms of architectural requirements. How does this transition impact the overall design and functionality of IoT systems?	10	

5. Attempt any *one* part of the following: 1 x 10 = 10

a.	Analyze the IoT reference model and its components. How do these components interact to support the functionality and scalability of IoT systems?	10	
b.	Describe the IoT Reference Architecture and its various views. How do these views contribute to a comprehensive understanding of IoT architecture?	10	

6. Attempt any *one* part of the following: 1 x 10 = 10

a.	Discuss the four aspects of business that need to be mastered to leverage IoT. How do these aspects influence the successful implementation of IoT solutions?	10	
b.	Evaluate the value creation potential of Big Data and Serialization in IoT. What are the key benefits and challenges associated with these technologies?	10	

7. Attempt any *one* part of the following: 1 x 10 = 10

a.	Analyze the importance of security, privacy, and trust in IoT data platforms for smart cities. How do these elements influence the success of smart city initiatives?	10	
b.	Describe the first steps towards developing a secure platform for IoT. What are the critical components and strategies involved in this process?	10	

QP24EP2_311 | 24-Jul-2024 1:37:11 PM | 115.245.67.130

QP24EP2_311 | 24-Jul-2024 1:37:11 PM | 115.245.67.130